

Sex Hormone-Binding Globulin (SHBG), estradiol and breast cancer.

The human serum Sex Hormone-Binding Globulin (SHBG) plays an important role in breast cancer pathophysiology and risk definition, since it regulates the bioavailable fraction of circulating estradiol. We here summarize data reported over the years concerning the involvement of SHBG and SHBG polymorphisms in the definition of breast cancer risk. We also report what is known about the direct action of SHBG in breast cancer cells, illustrating its interaction with these cells and the subsequent initiation of a specific intracellular pathway leading to cross-talk with the estradiol-activated pathway and, finally, to the inhibition of several effects of estradiol in breast cancer cells. In conclusion, as a result of its unique property of regulating the estrogen free fraction and cross-talking with the estradiol pathways, by inhibiting estradiol-induced breast cancer cell growth and proliferation, SHBG is associated with a reduced risk of developing the neoplasm after estrogen exposure. 2009 Elsevier Ireland Ltd. All rights reserved.